Ligjerata 8a

Fluidet

Shtypja atmosferike

$p = \rho g h = 13595 \frac{kg}{m^3} \cdot 9{,}81 \frac{m}{s^2} \cdot 0{,}76\,m = 101388{,}88\,Pa$

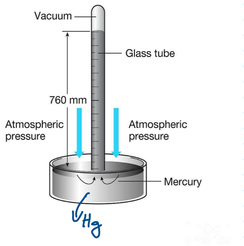
$\rho = 13595 \frac{kg}{m^3}$

$g = 9{,}81 \frac{m}{s^2}$

$h = 760\,mm = 0{,}76\,m$

$kg \frac{m}{s^2} \cdot \frac{1}{m^2} = \frac{N}{m^2} = Pa$

$p_{at} \approx 101325\,Pa$



$h \uparrow \quad p_a \downarrow$

$p = p(h)$

$dp = -\rho g\, dh$

$T = konst.$

$p \cdot V = p_0 V_0 \;/:$

$\frac{p \cdot V}{m} = \frac{p_0 V_0}{m_0} \qquad \rho = \frac{m}{V}$

$p \cdot \frac{1}{\frac{m}{V}} = p_0 \frac{1}{\frac{m_0}{V_0}} \Rightarrow \frac{p}{\rho} = \frac{p_0}{\rho_0} \to \rho = \frac{p \cdot \rho_0}{p_0}$

$dp = -\frac{p \cdot \rho_0}{p_0} g\, dh \;/\int$

$\int_{p_0}^{p} \frac{dp}{p} = -\int_0^h \frac{\rho_0}{p_0} g\, dh$

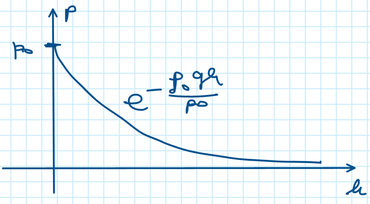
$\ln p \Big|_{p_0}^{p} = -\frac{\rho_0}{p_0} g \cdot h$

$\ln p - \ln p_0 = -\frac{\rho_0}{p_0} g h$

$\ln \frac{p}{p_0} = -\frac{\rho_0 g h}{p_0} \;/e$

$\frac{p}{p_0} = e^{-\frac{\rho_0 g h}{p_0}}$

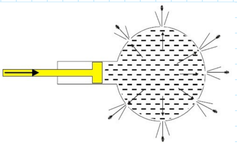
$p(h) = p_0 e^{-\frac{\rho_0 g h}{p_0}}$



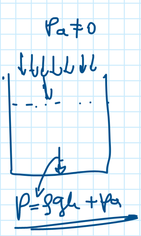
Përhapja e forcës në fluide
Ligji i Paskalit.

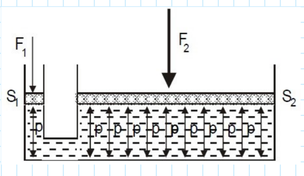
$p = \frac{F}{S} \quad ; \quad p = \frac{dF}{dS}$

$p = \frac{m \cdot g}{S} = \frac{\rho \cdot V \cdot g}{S} = \frac{\rho \cdot S \cdot h \cdot g}{S}$



$p = \rho g h$





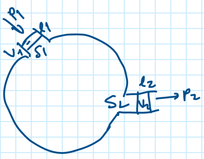
$F_2 = 100 F_1$

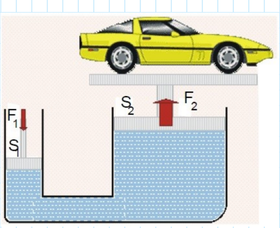
$S_2 = 100 S_1$

$p_1 = \frac{F_1}{S_1}$

$p_2 = \frac{F_2}{S_2} = \frac{100 F_1}{100 S_1} = \frac{F_1}{S_1} = p_1$

$p_2 = p_1$





$V_1 = V_2$

$V_1 = S_1 \cdot h_1$

$V_2 = S_2 \cdot h_2$

$S_1 h_1 = S_2 h_2$

$\frac{S_1}{S_2} = \frac{h_2}{h_1}$

$A_1 = F_1 \cdot h_1$

$A_2 = F_2 h_2$

$A_1 = A_2$

$F_1 h_1 = F_2 h_2$

$\frac{F_1}{F_2} = \frac{h_2}{h_1} = \frac{S_1}{S_2}$

$\frac{F_1}{F_2} = \frac{S_1}{S_2} \Rightarrow \frac{F_1}{S_1} = \frac{F_2}{S_2}$

$p_1 = p_2$